

Computer Skills

Chapter 3:

Numbers Systems & Conversions

University of Aden Faculty of Engineering
Information Technology (IT) –Computers Skills– First Semester

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Number Systems

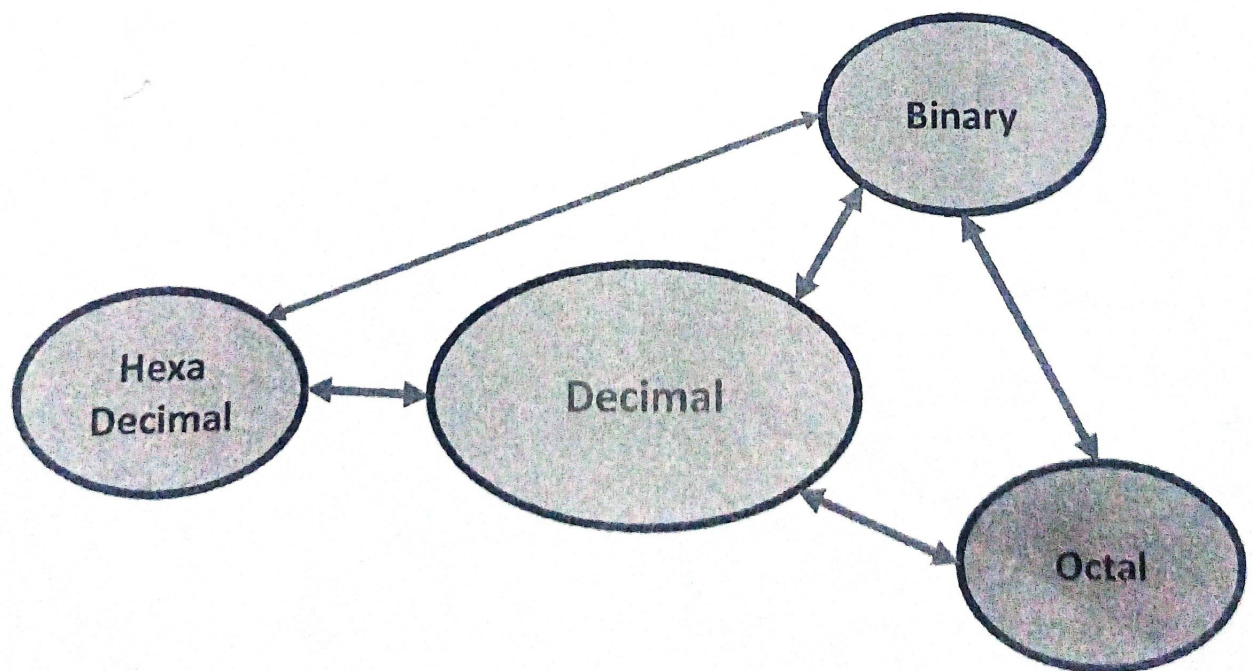
When we type some letters or words, the computer translates them in numbers as computers can understand only numbers. A computer can understand positional number system where there are only a few symbols called digits and these symbols represent different values depending on the position they occupy in the number.

Number System and Description :

- 1- Binary Number System Base 2. Digits used: 0, 1
- 2- Octal Number System Base 8. Digits used: 0 to 7
- 3- Decimal Number System Base 10. Digits used: 0 to 9
- 4- HexaDecimal Number System Base 16. Digits used: 0 to 9, Letters used: A- F

Binary Decimal Octal and Hexadecimal Conversion Chart

Octal	Decimal	Binary	Hexadecimal
0	0	0	0
1	1	0001	1
2	2	0010	2
3	3	0011	3
4	4	0100	4
5	5	0101	5
6	6	0110	6
7	7	0111	7
	8	1000	8
	9	1001	9
	10	1010	A
	11	1011	B
	12	1100	C
	13	1101	D
	14	1110	E
	15	1111	F



#Converting From any system to Decimal

Converting numbers from Binary System To Decimal:

a) $(110011)_2$

$$(1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1)_2 =$$

$$1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 =$$

$$32 + 16 + 0 + 0 + 2 + 1 = (51)_{10}$$

$$b) (1001)_2$$

$$\begin{aligned}
 (1 \quad 0 \quad 0 \quad 1)_2 &= \\
 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 &= \\
 8 + 0 + 0 + 1 &= (9)_{10}
 \end{aligned}$$

Converting numbers from Octal System To Decimal:

$$a) (275)_8$$

$$\begin{aligned}
 (2 \quad 7 \quad 5)_8 &= \\
 2 \times 8^2 + 7 \times 8^1 + 5 \times 8^0 &= \\
 128 + 56 + 5 &= (189)_{10}
 \end{aligned}$$

$$b) (3250)_8$$

$$\begin{aligned} (\quad 3 \quad \quad 2 \quad \quad 5 \quad \quad 0)_8 &= \\ 3 \times 8^3 + 2 \times 8^2 + 5 \times 8^1 + 0 \times 8^0 &= \\ 1536 + 128 + 40 + 0 &= (1704)_{10} \end{aligned}$$

Converting numbers from Hexadecimal System To Decimal:

2 8 10 12

$$a) (3B)_{16}$$

$$\begin{aligned} (\quad 3 \quad \quad B \quad)_{16} &= \\ 3 \times 16^1 + 11 \times 16^0 &= \\ 48 + 11 &= (59)_{10} \end{aligned}$$

b) $(7A9)_{16}$

$$\begin{aligned} & (\quad 7 \quad \quad A \quad \quad 9 \quad)_{16} = \\ & \quad 7 \times 16^2 + 10 \times 16^1 + 9 \times 16^0 = \\ & \quad 1792 + 160 + 9 = (1961)_{10} \end{aligned}$$

**2#Converting From
Decimal to any system**

Converting Numbers From Decimal To Binary:

Convert 14 to its binary equivalent.

Solution:

Step	Divided	Quotient	0 - r Remainder
1	$14/2$	$= 7$	
2	$7/2$	$= 3$	0
3	$3/2$	$= 1$	1
4	$1/2$	$= 0$	1

Therefore, $(14)_{10}$ is equivalent to $(1110)_2$

Convert 13_{10} to binary:

Division by 2	Quotient	Remainder
$13/2$	6	1
$6/2$	3	0
$3/2$	1	1
$1/2$	0	1

So $13_{10} = 1101_2$

Convert 174_{10} to binary:

Division by 2	Quotient	Remainder
$174/2$	87	0
$87/2$	43	1
$43/2$	21	1
$21/2$	10	1
$10/2$	5	0
$5/2$	2	1
$2/2$	1	0
$1/2$	0	1

So $174_{10} = 10101110_2$

Converting Numbers From Decimal To Octal :

1. Convert Decimal $(4321)_{10}$ to Octal

$$\begin{array}{rcl}
 4321/8 = 540 & \dots\dots\dots & 1 \\
 540/8 = 67 & \dots\dots\dots & 4 \\
 67/8 = 8 & \dots\dots\dots & 3 \\
 8/8 = 1 & \dots\dots\dots & 0 \\
 1 & \dots\dots\dots & 1
 \end{array}$$

↑
15 70

2. Convert Decimal $(35)_{10}$ to Octal

Division by 8	Quotient	Remainder (Octal)
35/8	4	3
4/8	0	4

The equivalent octal value is $(10341)_8$ The equivalent octal value is $(43)_8$

3. $(408)_{10} \longrightarrow (630)_8$

0-7

Step	Divided	Quotient	Remainder
1	408/8 =	51	0
2	51/8 =	6	3
3	6/8 =	0	6

$(630)_8$

4. $(175)_{10} \longrightarrow (257)_8$

Step	Divided	Quotient	Remainder
1	175/8 =	21	7
2	21/8 =	2	5
3	2/8 =	0	2

$(257)_8$

19 Numbers From Decimal To Hexadecimal :

Convert 7562_{10} to hex:

Division by 16	Quotient	Remainder (decimal)	Remainder (hex)
$7562/16$	472	10	A
$472/16$	29	8	8
$29/16$	1	13	D
$1/16$	0	1	1

So $7562_{10} = 1D8A_{16}$

Convert 35631_{10} to hex:

Division by 16	Quotient	Remainder (decimal)	Remainder (hex)
$35631/16$	2226	15	F
$2226/16$	139	2	2
$139/16$	8	11	B
$8/16$	0	8	8

So $35631_{10} = 8B2F_{16}$

$(445)_{10} \longrightarrow (1BD)_{16}$

Step	Divided	Quotient	Remainder
1	$445/16 =$	27	D
2	$27/16 =$	1	B
3	$1/16 =$	0	1



3#Converting From Binary to Octal and Hexadecimal

Conversion Of Numbers From Binary To Octal;

1. $(11010110)_2 \longrightarrow (326)_8$

0 1 1	0 1 0	1 1 0
4 2 1	4 2 1	4 2 1
$0+2+1=3$	$0+2+0=2$	$4+2+0=6$
3	2	6

11010110
011 010 110
421 421 421
0+2+1=3 0+2+0=2 4+2+0=6
3 2 6

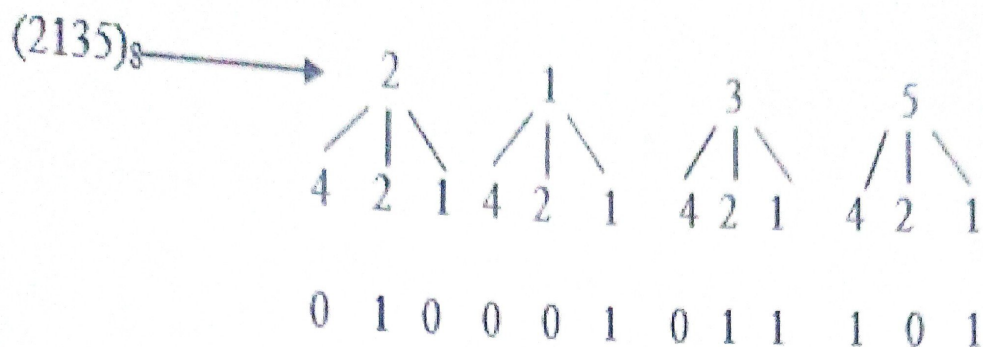
2. $(1001011010)_2 \longrightarrow (1132)_8$

0 0 1	0 0 1	0 1 1	0 1 0
4 2 1	4 2 1	4 2 1	4 2 1
$0+0+1=1$	$0+0+1=1$	$0+2+1=3$	$0+2+0=2$
1	1	3	2

3. $(11100110)_2 \longrightarrow (346)_8$

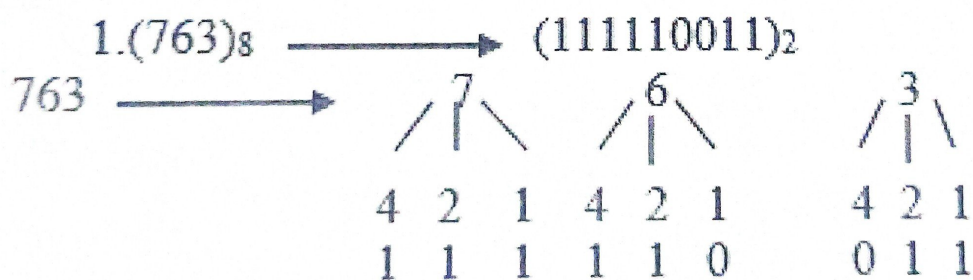
0 1 1	1 0 0	1 1 0
4 2 1	4 2 1	4 2 1
$0+2+1=3$	$4+0+0=4$	$4+2+0=6$
3	4	6

Conversion Of Numbers From Octal To Binary :

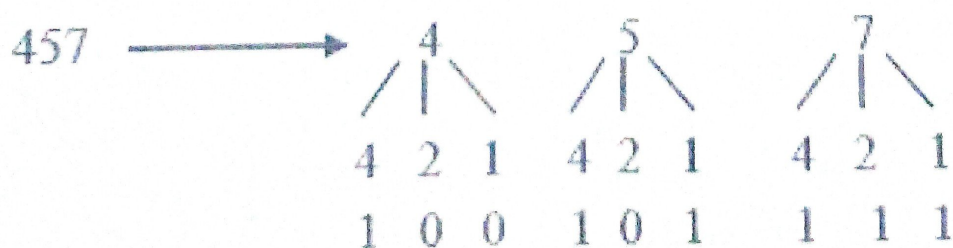


$(2135)_8 \longrightarrow (010001011101)_2$

2135
2 1 3 5
01 00 01 11 01
(010 001 011 101)₂



2. $(457)_8 \longrightarrow (100101111)_2$



Conversion Of Numbers From Binary , Hexadecimal:

$$\begin{array}{rcl}
 (111001011101)_2 & \longrightarrow & \begin{array}{ccc} 1110 & 0101 & 1101 \\ 8421 & 8421 & 8421 \\ 8+4+2+0=14 & 0+4+0+1=5 & 8+4+0+1=13 \\ E & 5 & D \end{array}
 \end{array}$$

$$(111001011101)_2 \longrightarrow (E5D)_{16}$$

Example:

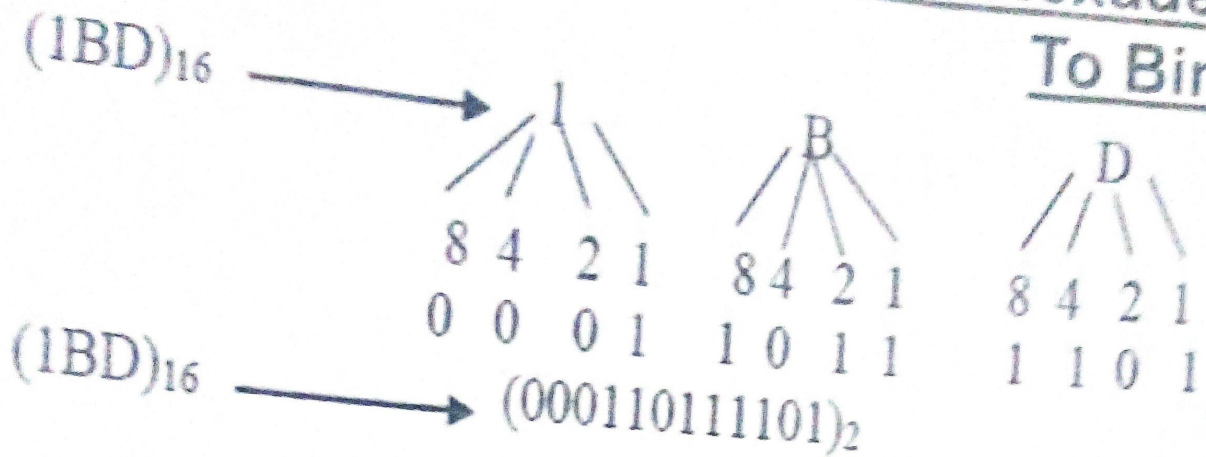
1. $(1001110011)_2$
2. $(111000110)_2$

Solution:

$$\begin{array}{rcl}
 1. (1001110011)_2 & \longrightarrow & (273)_{16} \\
 1001110011 & \longrightarrow & \begin{array}{ccc} 0010 & 0111 & 0011 \\ 8421 & 8421 & 8421 \\ 0+0+2+0=2 & 0+4+2+1=7 & 0+0+2+1=3 \\ 2 & 7 & 3 \end{array}
 \end{array}$$

$$\begin{array}{rcl}
 2. (111000110)_2 & \longrightarrow & (1C6)_{16} \\
 111000110 & \longrightarrow & \begin{array}{ccc} 0001 & 1100 & 0110 \\ 8421 & 8421 & 8421 \\ 0+0+0+1=1 & 8+4+0+0=12 & 0+4+2+0=6 \\ 1 & C & 6 \end{array}
 \end{array}$$

Conversion Of Numbers From Hexadecimal To Binary :



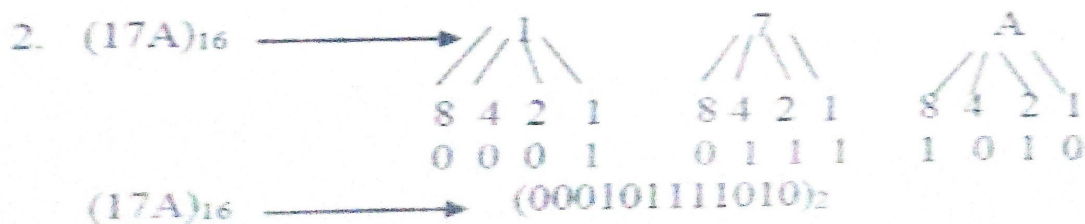
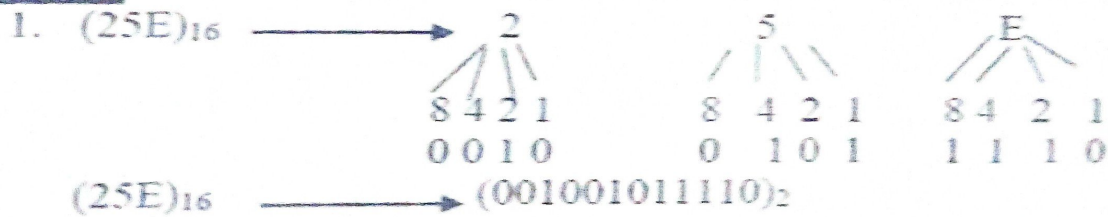
Handwritten notes: 8421, 8421, 8421

Handwritten binary: 000110111101

Example:

1. $(25E)_{16}$
2. $(17A)_{16}$

Solution:



4#Converting Between Octal and Hexadecimal

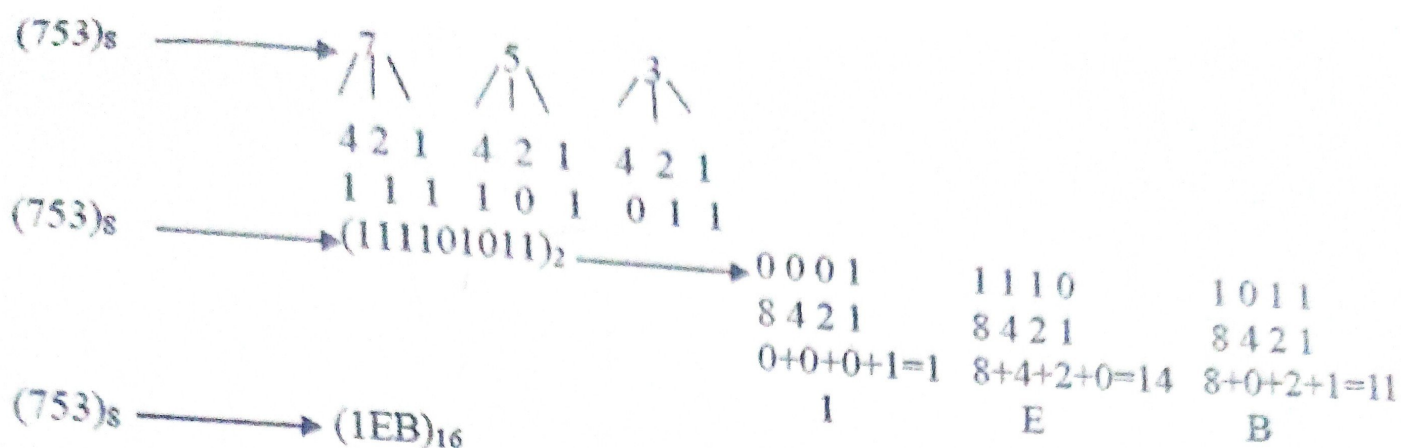
Conversion Of Numbers From Octal To Hexadecimal:

In order to convert a number from octal to hexadecimal system we need first to convert a number from octal to binary system and then straight to hexadecimal system.

$$(3457)_8 \longrightarrow \begin{array}{c} 3 \\ \swarrow \downarrow \searrow \\ 4 \ 2 \ 1 \\ 0 \ 1 \ 1 \end{array} \quad \begin{array}{c} 4 \\ \swarrow \downarrow \searrow \\ 4 \ 2 \ 1 \\ 1 \ 0 \ 0 \end{array} \quad \begin{array}{c} 5 \\ \swarrow \downarrow \searrow \\ 4 \ 2 \ 1 \\ 1 \ 0 \ 1 \end{array} \quad \begin{array}{c} 7 \\ \swarrow \downarrow \searrow \\ 4 \ 2 \ 1 \\ 1 \ 1 \ 1 \end{array}$$

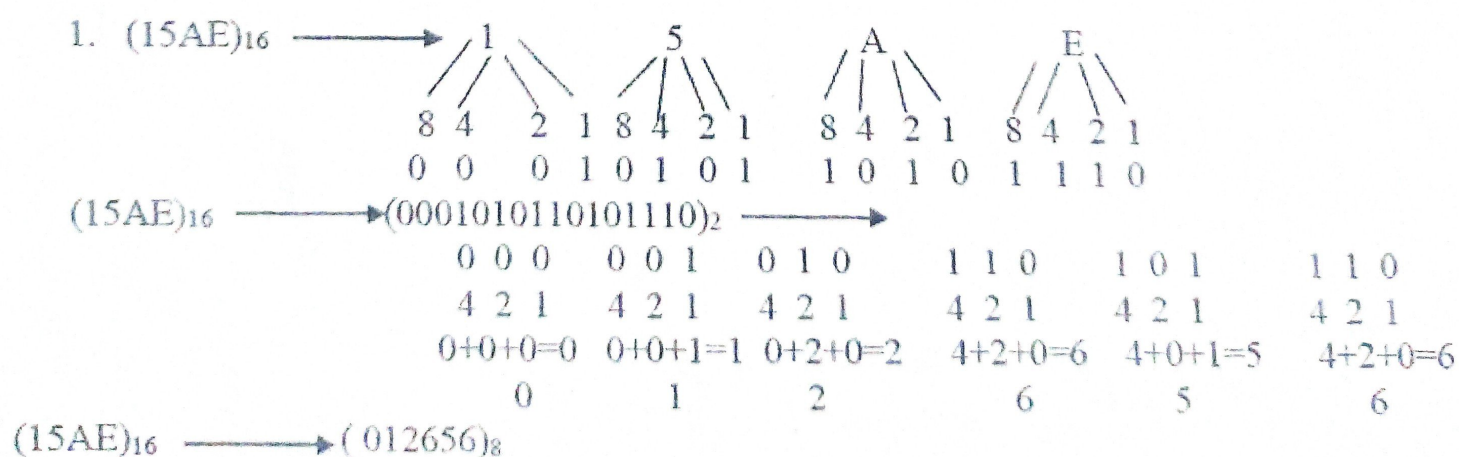
$$(3457)_8 \longrightarrow (011100101111)_2 \longrightarrow \begin{array}{ccc} 0111 & 0010 & 1111 \\ 8421 & 8421 & 8421 \\ 0+4+2+1=7 & 0+0+2+0=2 & 8+4+2+1=15 \\ 7 & 2 & F \end{array}$$

$$(3457)_8 \longrightarrow (72F)_{16}$$



Conversion Of Numbers From Hexadecimal To Octal:

In order to convert a number from hexadecimal to octal system we need first to convert a number from hexadecimal to binary system and then straight to octal system.



2. $(24C)_{16} \longrightarrow$

2			
8	4	2	1
0	0	1	0

4			
8	4	2	1
0	1	0	0

C			
8	4	2	1
1	1	0	0

$(24C)_{16} \longrightarrow (001001001100)_2 \longrightarrow$

0	0	1
4	2	1
$0+0+1=1$		
1		

0	0	1
4	2	1
$0+0+1=1$		
1		

0	0	1
4	2	1
$0+0+1=1$		
1		

1	0	0
4	2	1
$4+0+0=4$		
4		

$(24C)_{16} \longrightarrow (1114)_8$

Hexadecimal to Octal

$(87AD)_{16} \longrightarrow (1000\ 0111\ 1010\ 1101)_2 \longrightarrow (10365)$

8	7	A	D
8421	8421	8421	8421
1000	0111	1010	1101

Numbers System Conversion Web

<https://www.rapidtables.com/convert/number/base-converter.html>

001	000	011	110	101	101
421	421	421	421	421	421
001	000	011	110	101	101
1	0	3	6	5	5